

A) Control – Run Order TIC from Raw File

Blank n=314 | Check n=62 | ISTD n=81 | QC n=18 | Sample n=394 | QC TIC CV=7.5%



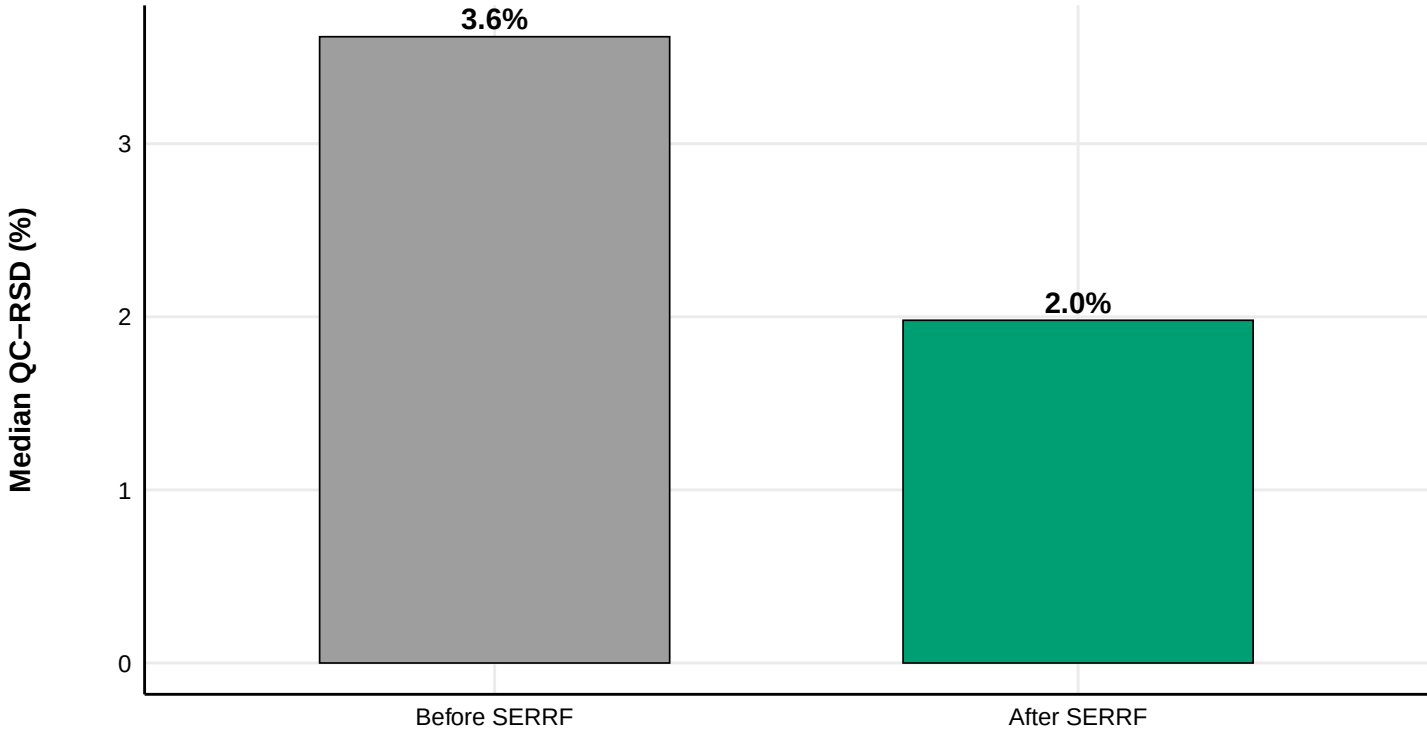
B) LowInput – Run Order TIC from Raw File

Blank n=289 | Check n=34 | ISTD n=74 | QC n=17 | Sample n=363 | QC TIC CV=6.0%



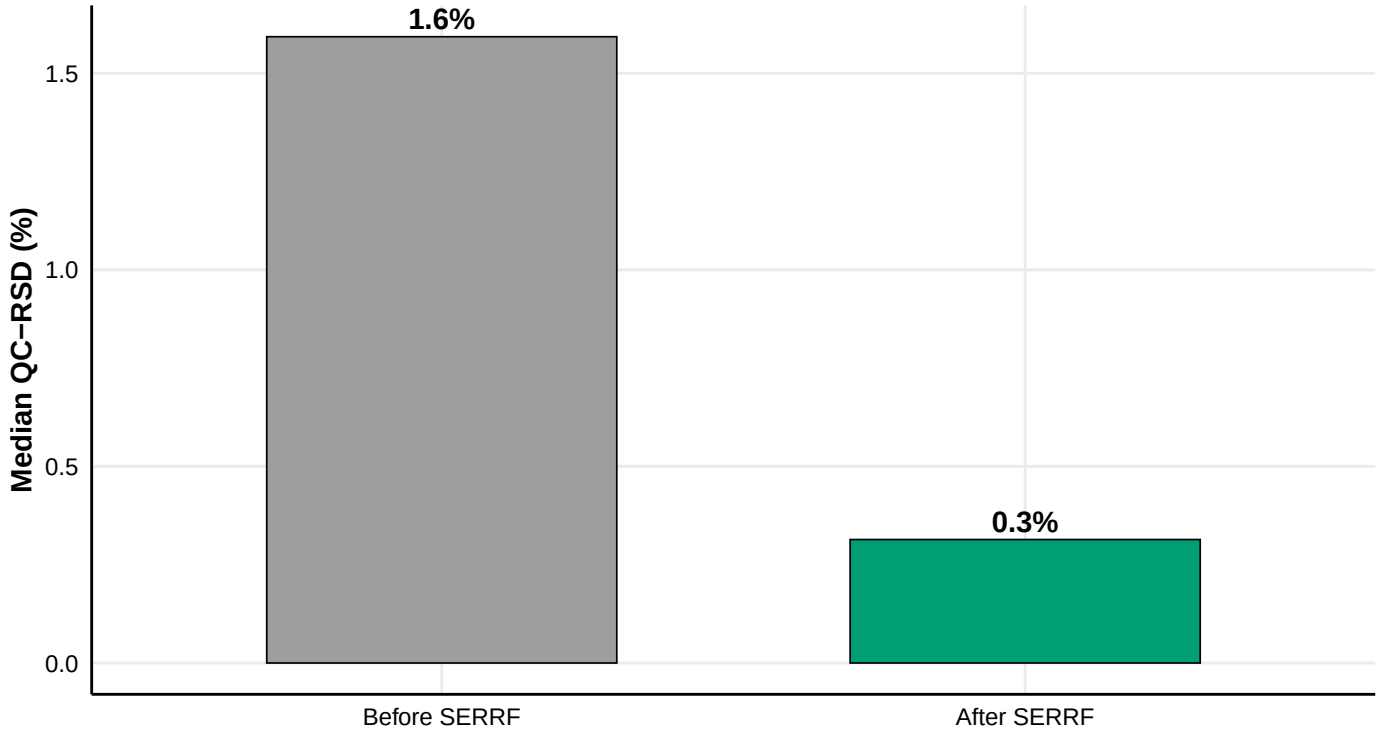
C) Control: QC–RSD Improvement After SERRF

QC–RSD < 30%: 75.7% ... 94.1% | Median drop: 45.3%



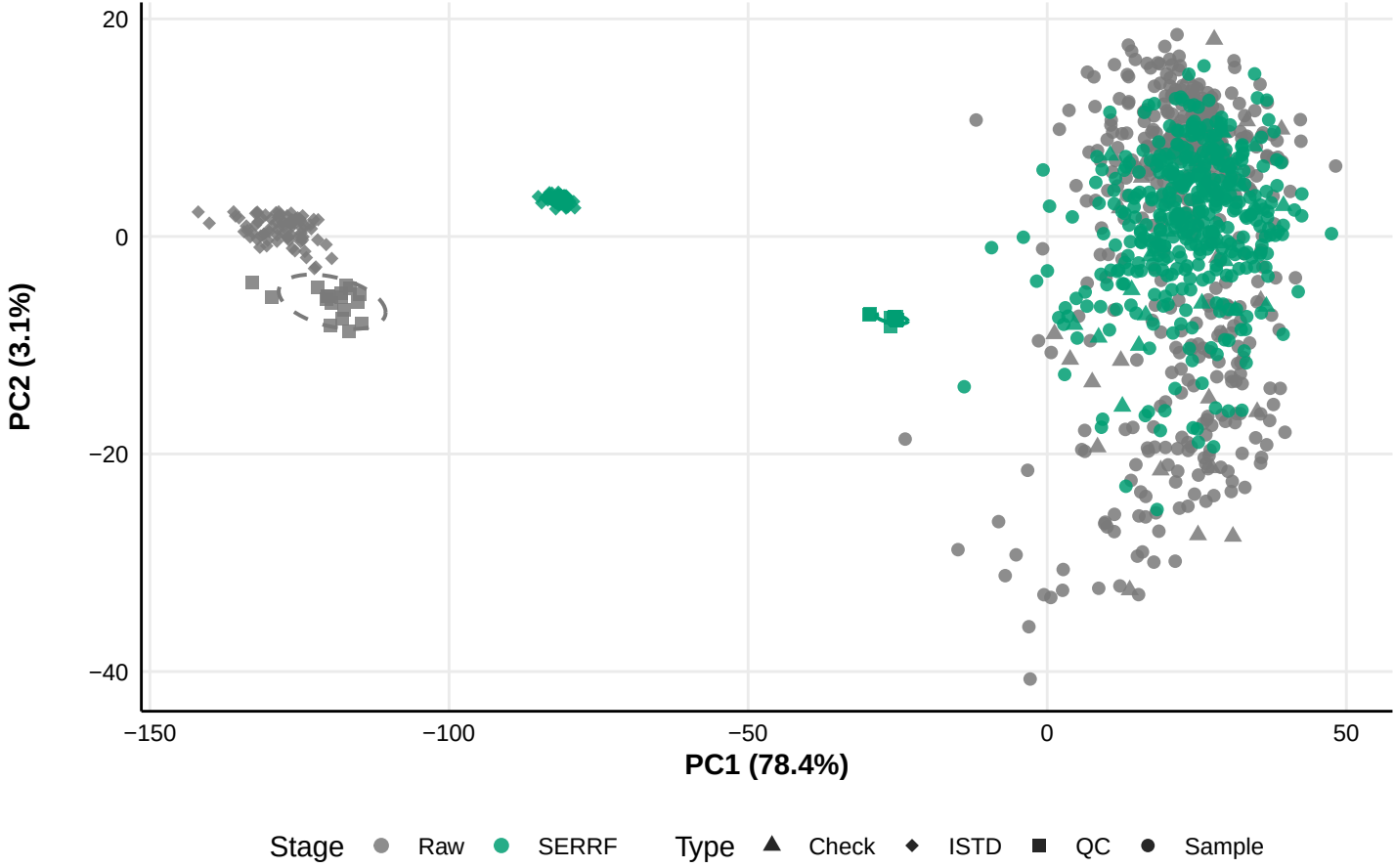
D) LowInput: QC–RSD Improvement After SERRF

QC–RSD < 30%: 77.3% ... 89.3% | Median drop: 80.3%



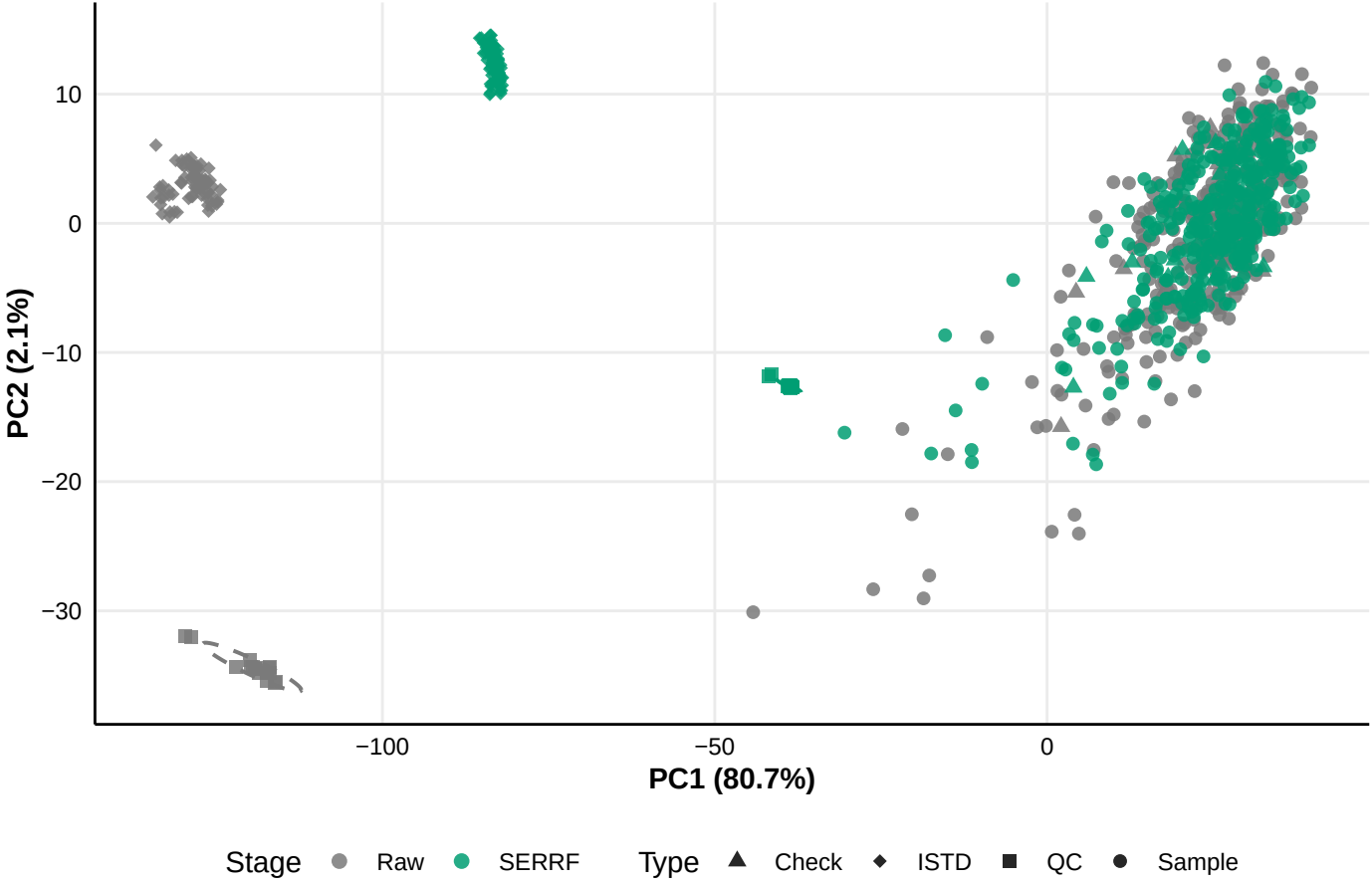
E) Control PCA: Raw vs SERRF

Matched injections: 524 | matched features: 3244 | dashed ellipse = QC cluster



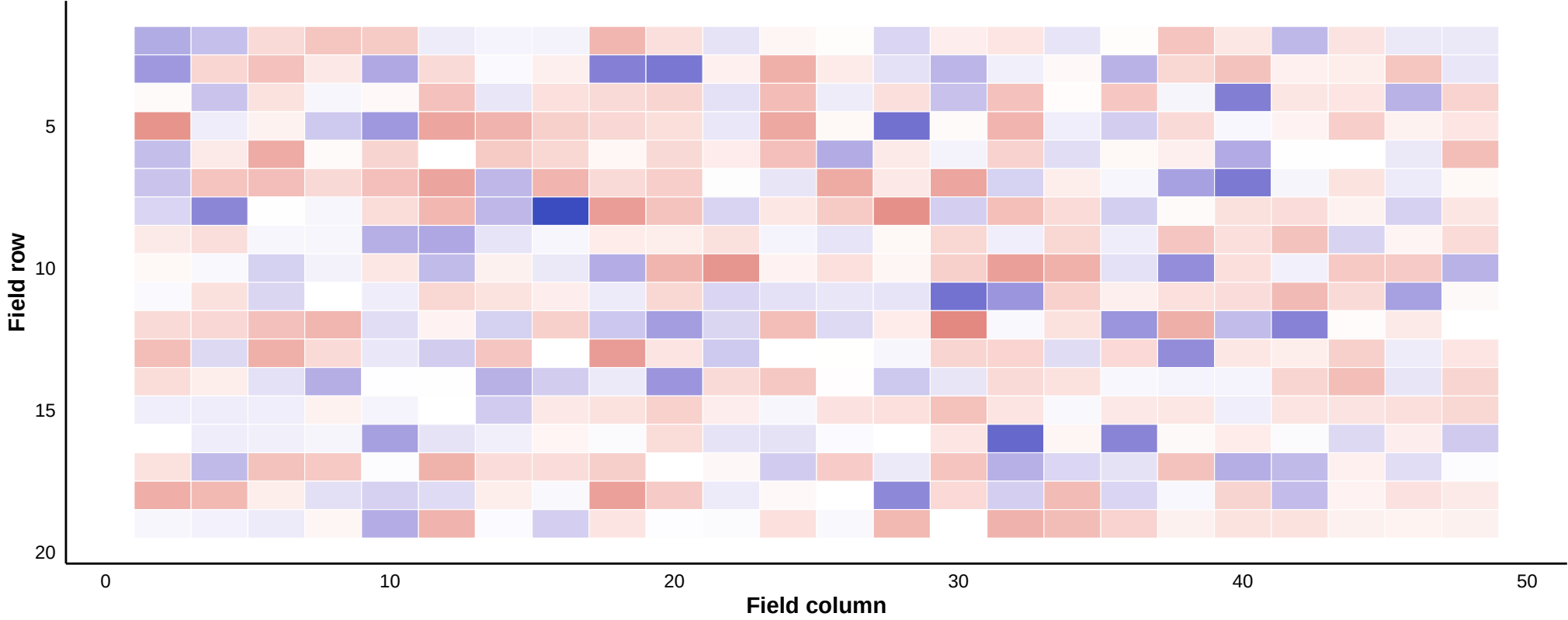
F) LowInput PCA: Raw vs SERRF

Matched injections: 471 | matched features: 3328 | dashed ellipse = QC cluster



G) Control SpATS Residual Map

Lipid: MGDG(18:3/18:3) | residuals after SpATS spatial fit



H) LowInput SpATS Residual Map

Lipid: MGDG(18:3/18:3) | residuals after SpATS spatial fit

